

QUESTION BANK

Based strictly on the textbook chapter (Ganita Manjari, Grade 9, Part II).

Purpose: This 100-mark practice bank is designed for high exam coverage. It deliberately repeats the chapter's core patterns: average of averages, weighted mean, mixtures/concentrations, custom weights, stacked bar charts, 100% stacked bar charts, and inference from visualisations. No question is intended to require content outside the uploaded chapter.

Important: No question bank can guarantee a 70% score because an actual paper can vary. However, if you can solve these questions correctly and understand the methods behind them, you will have practised the chapter's main question types very thoroughly.

Section	Questions	Marks
A. Warm-up / Easy	3×2	6
B. Normal / Standard	10×4	40
C. Competency-based	10×4	40
D. Case Studies	2×7	14
Total	25 questions	100

This question bank does not follow the CBSE's question grading pattern. 😂😂😂👉👉👉

Suggested attempt order: Warm-up → Normal → Competency based → Case Studies. This lets you secure the most direct marks first.

A. WARM-UP / EASY QUESTIONS — 6 MARKS

1. Find the combined average of two groups: Group A has average 18 with 5 values, and Group B has average 24 with 3 values. [2 marks]

*Note: for any dumb*** nerd values are student number.*

2. Two equal quantities of a drink have concentrations of 10% and 20% of a substance. What is the concentration after mixing them? Give a reason. [2 marks]

3. State one important difference between a stacked bar chart and a 100% stacked bar chart. [2 marks]

B. NORMAL / STANDARD QUESTIONS — 40 MARKS

4. Section A has 30 students with an average score of 72. Section B has 25 students with an average score of 76. Find the combined average score. [4 marks]

5. Three groups have averages 12, 15 and 18, with 5, 10 and 15 values respectively. Find the combined average. [4 marks]

6. A 500 mL lemonade has 10% jaggery and a 200 mL lemonade has 20% jaggery. Find the concentration of jaggery in the mixture. [4 marks]

7. A brass batch of 200 kg is 70% copper and another batch of 120 kg is 50% copper. If a third batch is 45% copper and the final mixture is 55% copper, find the weight of the third batch. [4 marks]

8. A student's marks are 60%, 64% and 73% in three components having weights 3:2:5. Find the weighted annual percentage. [4 marks]

9. A hotel rating is based on Food, Service and Ambience with weights 4:3:2. A customer gives ratings 5, 3 and 4 respectively. Find the combined rating. [4 marks]

10. A shop earns 30% profit on ₹8,000 worth of book sales and 50% profit on ₹1,000 worth of book-cover sales. Find the percentage profit on total sales. [4 marks]

11. A white stork travelled an average of 44.5 km per day for 20 days. On the 21st day it travelled 55 km. Find its average daily distance over all 21 days. [4 marks]

12. Using the chapter's family-expenditure data (Family A: Housing ₹1940, Food ₹1700, Education ₹1280, Transportation ₹1200, Healthcare ₹1770, Recreation ₹535), find Family A's total expenditure and the approximate percentage spent on housing. [4 marks]

13. Explain why the simple average of two group averages is not generally the correct combined average. Give a small numerical example in your explanation. [4 marks]

C. COMPETENCY-BASED QUESTIONS — 40 MARKS

14. A school has 40 students in Class A with an average height of 150 cm and 20 students in Class B with an average height of 156 cm. A student says the combined average must be 153 cm. Is the student correct? Calculate and justify. [4 marks]
15. Three weekly datasets have averages 12.8, 15.8 and 18 minutes. Each week contains 5 days. Explain why taking the simple average of the three weekly averages gives the correct overall average, and calculate it. [4 marks]
16. Two mixtures contain the same concentration of a substance: 5% in 300 mL and 5% in 700 mL. A student claims the mixture after combining them will have 10% concentration because $5 + 5 = 10$. Identify the error and find the correct concentration. [4 marks]
17. Mixture A contains 5% salt in 600 mL and Mixture B contains 8% sugar in 300 mL. Determine the concentration of salt and sugar in the final mixture, treating each substance separately. Explain why the answer cannot be found by simply averaging 5% and 8%. [4 marks]
18. A final assessment uses weights 4:5:6 for Strength, Flexibility and Agility. Rashi scores 60, 65, 70; Keerthi scores 55, 65, 75. Without doing the full calculation first, predict whose weighted score is higher. Then calculate both and verify your prediction. [4 marks]
19. A 100% stacked bar chart shows that Garden P has 50%, 30%, 20% of its annual flowers blooming in Summer, Monsoon and Winter, while Garden Q has 40%, 35%, 25%. Can you conclude that P had more summer flowers than Q? Explain why or why not. [4 marks]
20. A stacked bar chart is used for three families' monthly expenses. The purpose is to compare total spending as well as the components making up each total. Explain why a stacked bar chart is more suitable than a 100% stacked bar chart for this purpose. [4 marks]
21. The electricity-use data for 2005 is 360, 300, 120 and 220 units for Lighting, Cooling, Kitchen appliances and Other appliances. Find the percentage share of each category and explain what the corresponding 100% stacked bar would show. [4 marks]
22. A 100% stacked bar chart shows two schools with the same-looking 100% bars, but their actual numbers of students are unknown. A student concludes that both schools have the same number of students. Is this inference valid? Give a clear reason and, if useful, a numerical illustration. [4 marks]
23. A weighted average has values 10, 20 and 30 with weights 2, 3 and 5. (a) Find the weighted average. (b) If every weight is doubled, what happens to the weighted average? Justify algebraically. [4 marks]

D. CASE STUDIES — 14 MARKS

Case Study 1: School Assessment

A school calculates a student's final Mathematics result from three components: Internal Tests, Project and Final Exam. The marks are 68%, 72% and 80%, and the components have relative weights 3:2:5.

- (a) Why would a simple average of 68%, 72% and 80% not represent the school's stated weighting? [2 marks]
- (b) Calculate the student's weighted final percentage. [3 marks]
- (c) If the three components had equal weights instead, would the final percentage increase or decrease? Calculate the equal-weight average and compare. [2 marks]

Case Study 2: Family Expenditure and Visualisation

Three families record their average weekly expenditure (₹) in six categories. For Family C the amounts are: Housing 950, Food 1700, Education 1540, Transportation 1400, Healthcare 1300 and Recreation 150. The school wants to compare both total expenditure and the share of each category.

- (a) Find Family C's total expenditure. [2 marks]
- (b) Find the approximate percentage of Family C's expenditure spent on housing. [2 marks]
- (c) Which type of chart is more useful for comparing the total expenditures of families: a normal stacked bar chart or a 100% stacked bar chart? Give a reason. [2 marks]
- (d) Which chart is more useful for comparing the proportion of each category within families? State the chart and explain. [1 mark]

HIGH-YIELD REVISION CHECKLIST

- 1. Average of averages: Never give equal importance to group averages unless the groups have equal sizes.
- 2. Two groups: Combined average = $(a \times n + b \times m) / (n + m)$.
- 3. Three groups: Combined average = $(a \times p + b \times q + c \times r) / (p + q + r)$.
- 4. Mixtures: Think in terms of amount of substance $\div$ total mixture; larger quantities have greater influence.
- 5. Weighted mean: Weighted average = $\Sigma(\text{weight} \times \text{value}) / \Sigma \text{weights}$.
- 6. Custom weights: Convert marks to percentages first when the components have different maximum marks, then apply the stated weights.
- 7. Stacked bar chart: Useful when totals and their components both matter.
- 8. 100% stacked bar chart: Each bar has the same total length; it compares proportions/percentages, not absolute totals.
- 9. Inference: A higher percentage in a 100% stacked bar does not automatically mean a higher actual number.
- 10. Visualisation choice: Choose the chart according to what you need to compare—absolute amounts, totals, or proportions.